UNIT 8 · STANDARD 6.AT.8

Equations and Inequalities Problem Solving

Lesson 8-7

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can model and solve real-world problems using equations and inequalities.

Language Objective: I can explain my reasoning using the words model, equation, inequality, and reasonableness.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Model **Equation** **Inequality** **Reasonableness** **Variable** **Constraint**

Level 3: try the blanks from memory first, then check the bank.

1. A drawing, equation, or diagram that represents a situation is a _____.

2. A math sentence with an equal sign is an _____.

3. A math sentence that compares values with $<$, $>$, $\leq$, or $\geq$ is an _____.

4 Checking whether an answer makes sense for the problem is checking its _____ .

5 A letter that stands for the unknown amount in a problem is a _____ .

6 A limit that tells which values are allowed in a problem is a _____ .

Write About the Math

How much more can the detective spend? How much does each flashlight cost? When do you use an equation versus an inequality?
¿Cuánto más puede gastar la detective? ¿Cuánto cuesta cada linterna? ¿Cuándo usas una ecuación y cuándo una desigualdad?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Model** *Modelar*
- ☐ **Equation** *Ecuación*
- ☐ **Inequality** *Desigualdad*
- ☐ **Reasonableness** *Razonabilidad*
- ☐ **Variable** *Variable*

Model: *An exact total points to an equation; a limit or range points to an inequality.* — Students use signal words ('equals/total' for an equation; 'at least/at most/no more than' for an inequality) to choose the correct model.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can spend at most $__ more because r ≤ 500 − ___ = ___. Each flashlight costs $__ because f = 60 ÷ ___ = ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.